STAT 516 - Homework 5 Answer Key

1. **(8 points)** Let X and Y be continuous random variables with joint density function

$$f(x,y) = \begin{cases} kx^{-2}e^{-y/2}, & x > 1,\ y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find the value of k.

Since $f(x,y)$ is a joint density function,

$$\int_1^\infty \int_0^\infty kx^{-2}e^{-y/2}\,dy\,dx = 1.$$

$$k\left(\int_1^\infty x^{-2}\,dx\right)\left(\int_0^\infty e^{-y/2}\,dy\right) = 1.$$

$$k(1)(2) = 1.$$

Thus,

$$k = \frac{1}{2}.$$

2. **(8 points)** Let X and Y be discrete random variables with joint probability function

$$f(x,y) = \begin{cases} \frac{1}{4}, & x = 1,2 \text{ and } y = 1,2, \\ 0, & \text{otherwise.} \end{cases}$$

Let $U = X + Y$. Find the probability function of U.

The possible values of $U = X + Y$ are $2, 3, 4$.

$$P(U=2) = P(X=1, Y=1) = \frac{1}{4}.$$

$$P(U=3) = P(X=1, Y=2) + P(X=2, Y=1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

$$P(U=4) = P(X=2, Y=2) = \frac{1}{4}.$$

Therefore,

$$P(U=u) = \begin{cases} \frac{1}{4}, & u = 2, \\ \frac{1}{2}, & u = 3, \\ \frac{1}{4}, & u = 4, \\ 0, & \text{otherwise.} \end{cases}$$

3. **(8 points)** A policy pays 200 dollars for each lost workday and 100 dollars for each doctor visit. Let X be the number of lost workdays and Y be the number of doctor visits. The joint probability function of X and Y is given by

$X\backslash Y$	0	1	2
0	0.20	0.10	0.05
1	0.08	0.12	0.07
2	0.06	0.10	0.08
3	0.04	0.06	0.04

Calculate the probability that the policy pays at least 500 dollars.

The policy payment is

$$200X + 100Y.$$

We want

$$P(200X + 100Y \geq 500).$$

For $X = 2$, we need $Y \geq 1$. This gives

$$P(X = 2, Y = 1) + P(X = 2, Y = 2) = 0.10 + 0.08.$$

For $X = 3$, all values of Y qualify. This gives

$$P(X = 3, Y = 0) + P(X = 3, Y = 1) + P(X = 3, Y = 2) = 0.04 + 0.06 + 0.04.$$

Thus,

$$P(200X + 100Y \geq 500) = 0.10 + 0.08 + 0.04 + 0.06 + 0.04 = 0.32.$$

4. **(10 points)** Let X and Y be continuous random variables with joint density function

$$f(x, y) = \begin{cases} \frac{1}{4}xy, & 0 < x < 4,\ 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$P\left(\frac{X}{4} < Y < \frac{X}{2}\right)$$

Hint: You may use $P\left(Y < \frac{X}{2}\right) - P\left(Y < \frac{X}{4}\right)$.

For $0 < x < 2$, the upper bound is $x/2$. For $2 < x < 4$, the upper bound is 1. Thus,

$$\begin{aligned}
P\left(\frac{X}{4} < Y < \frac{X}{2}\right) &= P\left(Y < \frac{X}{2}\right) - P\left(Y < \frac{X}{4}\right) \\
&= \int_0^2 \int_{x/4}^{x/2} \frac{1}{4}xy\,dy\,dx + \int_2^4 \int_{x/4}^{1} \frac{1}{4}xy\,dy\,dx \\
&= \frac{1}{4}\int_0^2 x\left[\frac{y^2}{2}\right]_{x/4}^{x/2} dx + \frac{1}{4}\int_2^4 x\left[\frac{y^2}{2}\right]_{x/4}^{1} dx \\
&= \frac{1}{8}\int_0^2 x\left(\frac{x^2}{4} - \frac{x^2}{16}\right) dx + \frac{1}{8}\int_2^4 x\left(1 - \frac{x^2}{16}\right) dx \\
&= \frac{1}{8}\int_0^2 \frac{3x^3}{16}\,dx + \frac{1}{8}\int_2^4 \left(x - \frac{x^3}{16}\right) dx \\
&= \frac{3}{128}\left[\frac{x^4}{4}\right]_0^2 + \frac{1}{8}\left[\frac{x^2}{2} - \frac{x^4}{64}\right]_2^4 \\
&= \frac{3}{32} + \frac{9}{32} \\
&= \frac{3}{8}
\end{aligned}$$

5. **(9 points)** Let X be the portion of an insurance claim representing damage to a building, and let Y be the portion representing damage to the remaining property. The joint density function is

$$f(x,y) = \begin{cases} 12(1-x-y)^2, & x > 0,\ y > 0,\ x+y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate the probability that the portion of the claim representing damage to the building is less than 0.25.

We want

$$P(X < 0.25) = \int_0^{0.25} \int_0^{1-x} 12(1-x-y)^2\,dy\,dx.$$

Let $u = 1 - x - y$. Then

$$\int_0^{1-x} 12(1-x-y)^2\,dy = -\int_{1-x}^{0} 12(u)^2\,du = 4(1-x)^3.$$

Thus,

$$P(X < 0.25) = \int_0^{0.25} 4(1-x)^3\,dx$$

$$= \left[-(1-x)^4\right]_0^{0.25} = 1 - \left(\frac{3}{4}\right)^4 = \frac{175}{256}.$$

6. **(9 points)** Let X and Y be random losses with joint density function

$$f(x, y) = 4e^{-2x-2y}, \quad x > 0,\ y > 0.$$

An insurance policy reimburses the amount $X + Y$. Calculate the probability that the reimbursement is less than 2.

We want

$$P(X + Y < 2) = \int_0^2 \int_0^{2-x} 4e^{-2x-2y}\, dy\, dx.$$

First integrate with respect to y:

$$\int_0^{2-x} 4e^{-2x-2y}\, dy = 4e^{-2x} \left[-\frac{1}{2}e^{-2y}\right]_0^{2-x}.$$

$$= 2e^{-2x}\left(1 - e^{-2(2-x)}\right).$$

Thus,

$$\begin{aligned} P(X + Y < 2) &= \int_0^2 2e^{-2x}\left(1 - e^{-4+2x}\right) dx. \\ &= \int_0^2 2e^{-2x}\, dx - \int_0^2 2e^{-4}\, dx. \\ &= \left[-e^{-2x}\right]_0^2 - 4e^{-4}. \\ &= 1 - e^{-4} - 4e^{-4} \\ &= 1 - 5e^{-4}. \end{aligned}$$

7. **(10 points)** A device stops running when the first of two components fails. The lifetimes of the two components, in hours, have joint density function

$$f(x, y) = \frac{x + 2y}{12}, \quad 0 < x < 2,\ 0 < y < 2.$$

Calculate the probability that the device fails during its first hour of operation, i.e., $P(\min(X, Y) < 1)$. (Hint: use the complement rule)

Use the complement:

$$P(\min(X, Y) < 1) = 1 - P(X \geq 1,\ Y \geq 1).$$

$$P(X \geq 1,\ Y \geq 1) = \int_1^2 \int_1^2 \frac{x + 2y}{12}\, dy\, dx.$$

$$= \frac{1}{12} \int_1^2 \int_1^2 (x + 2y)\, dy\, dx.$$

$$= \frac{1}{12}\left(\frac{3}{2}+3\right) = \frac{1}{12}\cdot\frac{9}{2} = \frac{3}{8}.$$

Therefore,

$$P(\min(X,Y) < 1) = 1 - \frac{3}{8} = \frac{5}{8}.$$

8. **(10 points)** Let X and Y have joint density function

$$f(x,y) = \begin{cases} \frac{1}{6}, & 0 < x < 3,\ x < y < x+2, \\ 0, & \text{otherwise.} \end{cases}$$

Determine the marginal density function of Y.

The length of the possible x-interval depends on y.

For $0 < y < 2$:

$$0 < x < y.$$

For $2 \le y < 3$:

$$y - 2 < x < y.$$

For $3 \le y < 5$:

$$y - 2 < x < 3.$$

Thus,

$$f_Y(y) = \begin{cases} \int_0^y \frac{1}{6}\,dx, & 0 < y < 2, \\ \int_{y-2}^y \frac{1}{6}\,dx, & 2 \le y < 3, \\ \int_{y-2}^3 \frac{1}{6}\,dx, & 3 \le y < 5, \\ 0, & \text{otherwise.} \end{cases}$$

So,

$$f_Y(y) = \begin{cases} \frac{y}{6}, & 0 < y < 2, \\ \frac{1}{3}, & 2 \le y < 3, \\ \frac{5-y}{6}, & 3 \le y < 5, \\ 0, & \text{otherwise.} \end{cases}$$

9. **(9 points)** Let X and Y be discrete random variables with joint probability function

$$f(x,y) = \begin{cases} \frac{x+y}{21}, & (x,y) = (1,1),(1,2),(2,1),(2,2),(3,1),(3,2), \\ 0, & \text{otherwise.} \end{cases}$$

Determine the marginal probability function of X.

The marginal probability function of X is

$$P(X=x)=\sum_y f(x,y).$$

For $x=1$,

$$P(X=1)=\frac{1+1}{21}+\frac{1+2}{21}=\frac{2}{21}+\frac{3}{21}=\frac{5}{21}.$$

For $x=2$,

$$P(X=2)=\frac{2+1}{21}+\frac{2+2}{21}=\frac{3}{21}+\frac{4}{21}=\frac{7}{21}=\frac{1}{3}.$$

For $x=3$,

$$P(X=3)=\frac{3+1}{21}+\frac{3+2}{21}=\frac{4}{21}+\frac{5}{21}=\frac{9}{21}=\frac{3}{7}.$$

Therefore,

$$P(X=x)=\begin{cases}\frac{5}{21}, & x=1,\\ \frac{1}{3}, & x=2,\\ \frac{3}{7}, & x=3,\\ 0, & \text{otherwise.}\end{cases}$$

10. **(10 points)** Let X and Y be continuous random variables with joint density function

$$f(x,y)=\begin{cases}\frac{2}{5}(x+y), & 0<x<1,\ x<y<2,\\ 0, & \text{otherwise.}\end{cases}$$

Find the marginal density function of Y.

If $0<y<1$, then

$$0<x<y.$$

If $1\le y<2$, then

$$0<x<1.$$

Therefore,

$$f_Y(y)=\begin{cases}\int_0^y \frac{2}{5}(x+y)\,dx, & 0<y<1,\\ \int_0^1 \frac{2}{5}(x+y)\,dx, & 1\le y<2,\\ 0, & \text{otherwise.}\end{cases}$$

For $0<y<1$,

$$f_Y(y)=\frac{2}{5}\left[\frac{x^2}{2}+yx\right]_0^y=\frac{2}{5}\left(\frac{y^2}{2}+y^2\right)=\frac{3y^2}{5}.$$

For $1 \le y < 2$,

$$f_Y(y) = \frac{2}{5}\left[\frac{x^2}{2} + yx\right]_0^1 = \frac{2}{5}\left(\frac{1}{2} + y\right) = \frac{2y+1}{5}.$$

Thus,

$$f_Y(y) = \begin{cases} \frac{3y^2}{5}, & 0 < y < 1, \\ \frac{2y+1}{5}, & 1 \le y < 2, \\ 0, & \text{otherwise.} \end{cases}$$

11. **(9 points)** Let X and Y have a uniform joint distribution over the region

$$\{(x, y) : x^2 + y^2 \le 4,\ x > 0,\ y > 0\}.$$

Find the marginal density function of X.

The region is a quarter disk with radius 2. Its area is

$$\frac{1}{4}\pi(2)^2 = \pi.$$

Since the joint distribution is uniform,

$$f(x, y) = \frac{1}{\pi}$$

on the quarter disk.

For a fixed x, y ranges from

$$0 < y < \sqrt{4 - x^2}.$$

Thus,

$$f_X(x) = \int_0^{\sqrt{4-x^2}} \frac{1}{\pi}\, dy.$$

$$f_X(x) = \frac{\sqrt{4 - x^2}}{\pi}, \quad 0 < x < 2.$$

Therefore,

$$f_X(x) = \begin{cases} \frac{\sqrt{4 - x^2}}{\pi}, & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$